Sprint Plan

ShopSense: the personalised shopping research agent

Document	Sprint Plan
Project	ShopSense: the personalised shopping research agent
Course	IT314 Software Engineering
Team	Group 13 (10 members)
Version	1.0
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This document sets out how ShopSense is delivered over six two-week sprints. For each sprint it gives the dates, the sprint goal, the user stories, the work of every role, the testing focus and what is shown at the sprint review.

1. The plan at a glance

Sprint	Theme	Planned dates	Review	Scrum Master	Story points
Sprint 1	The walking skeleton	21 Sep – 4 Oct	Sat 3 Oct	Jhanvi Barot	36
Sprint 2	Research and the safety net	5 – 18 Oct	Sat 17 Oct	Yashkumar Patel	37
Sprint 3	Trustworthy end to end	19 Oct – 1 Nov	Sat 31 Oct	Vaishvi Mehta	40
Sprint 4	Smarter and safer	2 – 15 Nov	Sat 14 Nov	Harshkumar Kodavala	36
Sprint 5	Feature-complete	16 – 29 Nov	Sat 28 Nov	Pal Jalodara	41 (36 core + 5 Plus)
Sprint 6	Worst-case proof and presentation	30 Nov – 13 Dec	Sat 12 Dec	Palak Patel	49 (31 core + 10 Plus + 8 Stretch)
Total		21 Sep – 13 Dec 2026			**239**

Dates are the planned baseline for 2026. If a sprint runs late, the later sprints move by the same amount; their order and owners stay the same. Core stories are committed. Plus stories are the first to move to a later sprint if a sprint runs late, and the Stretch story is done only if time allows.

2. Team and roles

Om Pandya is the Product Owner and group leader. The Scrum Master role rotates each sprint, as shown in the table above.

Member	Student ID	Role	Main responsibilities
Om Pandya	202401136	ML Engineer · Product Owner · Group Leader	Owns and orders the backlog, accepts stories at review, repo administration and merges; leads product matching, the adaptive interview and preference memory.
Rushiraj Navar	202401127	ML Engineer	Leads source trust, ranking confidence and the citation check; builds the evaluation runners and labelled sets for them.

Member	Student ID	Role	Main responsibilities
Jhanvi Barot	202401076	AI Engineer (LLM integration)	Owns the LLM wrapper and every language step: intent detection, criteria, claim extraction, explanations, the reply classifier and enrichment.
Harshkumar Kodavala	202401121	Data & Evaluation Engineer	Owns offline testing data, the recorded-run replay and quality gate, login and data isolation, the metrics report and the 20-brief study.
Palak Patel	202401133	QA Engineer	Owns CI hardening, golden tests, the worst-case suite, sprint review checks and the weekly citation audit.
Pal Jalodara	202401074	Backend Engineer	Built the walking skeleton; owns untrusted-content handling, the rubric, graceful failure handling, extra sources, caching and module interfaces.
Vaishvi Mehta	202401117	Backend Engineer	Owns the database and migrations, Reddit quality and counting, filtering and ranking, memory storage, data controls and rate limits.
Jenish Kavad	202401088	Backend Engineer	Owns the interview logic and pre-search API, research and parallel fetching, exclusions, the evidence API, prices and traceability.
Yashkumar Patel	202401109	Frontend Engineer	Owns the web app shell, request box, research progress, results page, login screens, compare and re-rank, and memory settings.
Manu Pardhan	202301020	Frontend Engineer & Testing	Owns the interview chat, rubric editor, confidence and explanation views, remembered answers, price cards and accessibility.

3. How the team works

Scrum events

- **Sprint planning.** At the start of each sprint, the team confirms the stories and splits them into tasks with one owner each.
- **Daily standup.** A short written update every day in the team channel: what was done, what is next and what is blocked.
- **Backlog refinement.** During each sprint, the Product Owner prepares and orders the stories for the next sprint.
- **Sprint review.** On the Saturday at the end of week two: a live demonstration and a live failure drill.
- **Retrospective.** After each review, the team records what to keep and what to change.

Types of work

Type	Meaning
Build	Code, tests and documentation, merged into the repository through a reviewed pull request.
Screen design	A screen is designed and approved by the team first; a separate Build task then builds it and connects it to the API.
Model training	A model is trained in a notebook from a script kept in the repository, then connected to its stage behind the existing fallback.
Research and process	Work done by hand: surveys, interviews, labelling, testing sessions and Scrum events.

Definition of done

- The acceptance criteria of the story pass.
- The work is merged through a pull request that another member has reviewed.
- Continuous integration (lint, type checks and tests) passes on the pull request.
- No API key, password or other secret is committed.
- Every stage keeps a working fallback, so a new feature can never stop a search from finishing.
- Work is committed and merged under the name of the member who did it.

4. Sprint 1: The walking skeleton

Planned dates	21 September – 4 October 2026
Sprint review	Saturday 3 October 2026
Scrum Master	Jhanvi Barot
Story points	36
Sprint goal	A request typed in the browser or the terminal runs through every stage and returns a ranked list, and the app understands what the shopper wants.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-01	Describe the need in plain words	Must	5	Yashkumar Patel, with Vaishvi Mehta
US-02	Detect category, use, budget and rules	Must	5	Jhanvi Barot
US-03	Generate category-specific criteria	Must	5	Jhanvi Barot
US-05	One question at a time, adapted to earlier answers	Must	8	Jenish Kavad (4 points), Manu Pardhan (4 points)
US-25	Repository, CI and secret scanning	Must	3	Palak Patel
US-26	One LLM wrapper with fallback and context fitting	Must	5	Jhanvi Barot
US-52	Walking skeleton, pipeline runner and CLI	Must	5	Pal Jalodara
	Total		**36**	

Work by role

AI Engineer

Task	Story	Type	Owner	Done when
T-03 LLM wrapper One wrapper over Groq, OpenRouter, Cerebras and Gemini (public text only), with a fallback chain, schema check, context fitting and a call log. *Starts after: T-02*	US-26	Build	Jhanvi Barot	With one provider key removed, a call still succeeds; tests use fake providers

Task	Story	Type	Owner	Done when
T-04 Intent detection Category, use, budget (k, lakh, ₹), region and rules into a validated Intent, with a keyword fallback when every LLM fails. *Starts after: T-03*	US-02	Build	Jhanvi Barot	“Gaming laptop under 80k” parses; “watch” asks which kind; with every LLM off, keywords still detect category and budget
T-05 Criteria generation 7 to 10 criteria per category with one-line definitions, value for money always included, one retry, then a fixed fallback set; saved per category. *Starts after: T-03*	US-03	Build	Jhanvi Barot	Three categories, one never seen, return criteria; the fallback set appears when generation fails
Ceremonies Daily written standup, refinement for Sprint 2, review and retrospective.	—	Research and process	Jhanvi Barot	Notes in #ceremonies
Survey, 30+ responses Google Form with the eight plan questions. *Starts at the start of the sprint*	—	Research and process	Jhanvi Barot	30 or more responses

Backend Engineer

Task	Story	Type	Owner	Done when
T-01 Walking skeleton Stage contracts, stub stages, pipeline runner with time limits and fallbacks, CLI. *Starts after: starter pack pushed*	US-52	Build	Pal Jalodara	CLI prints a ranked list from stub stages; every stage has a contract test
T-06 Interview logic and pre-search API Default-order questions, skipping related questions, going back to change an answer; API for intake → criteria → interview → rubric. *Starts after: T-04, T-05*	US-05	Build	Jenish Kavad	5+ questions run through the API and answers are stored; each turn under 5 s
T-14 Database start Supabase project, first migration (requests saved with a session ID), DB module that lets the pipeline run without a database. *Starts after: T-01*	US-01	Build	Vaishvi Mehta	Each request is saved with a session ID; with the DB down, a search still finishes
API sign-ups and terms check Every API smoke-tested; real limits recorded; Reddit and YouTube terms checked. *Starts at the start of the sprint*	—	Research and process	Jenish Kavad	docs/api-setup.md updated
Concept poster A1/A2 poster; Palak Patel co-owns it. *Starts at the start of the sprint*	—	Research and process	Vaishvi Mehta	Poster printed
Expert interview One person who knows laptops or phones; checked criteria list. *Starts at the start of the sprint*	—	Research and process	Pal Jalodara	Notes in #elicitation

Frontend Engineer

Task	Story	Type	Owner	Done when
T-07 Frontend setup Next.js app shell, TypeScript types generated from the contracts, API client, mock-data mode from search_result.json. *Starts after: T-02*	—	Build	Yashkumar Patel	App builds in CI; the plain results page renders the mock result
T-08 Request box design Request box, follow-up question with category buttons, example requests. *Starts after: T-07*	US-01	Screen design	Yashkumar Patel	Design approved by the team in #dev
T-09 Request box build and connection Build the approved design in the repository and connect it to the intake API; research never starts until the category is known. *Starts after: T-06, T-08*	US-01	Build	Yashkumar Patel	Empty, unclear and product-less requests each get the right follow-up

Frontend Engineer & Testing

Task	Story	Type	Owner	Done when
T-10 Interview chat design Chat window, one question per turn, quick-reply buttons, free text, edit an earlier answer. *Starts after: T-07*	US-05	Screen design	Manu Pardhan	Design approved by the team in #dev
T-11 Interview chat build and connection Rebuild the chat in the repo and connect it to the interview API. *Starts after: T-06, T-10*	US-05	Build	Manu Pardhan	A full interview runs in the browser and the answers are stored

ML Engineer

Task	Story	Type	Owner	Done when
T-12 ML workspace (Phase 0) ml/DATA.md and ml/MODELS.md, labelling CSV format for product pairs, label checker, Cohen’s kappa script. *Starts after: T-02*	—	Build	Om Pandya	Kappa script runs on a 50-pair sample; the raw data folder is git-ignored
T-13 Matching eval scaffold (Phase 0) evals/matching: pair-set loader and RapidFuzz baseline that prints precision, recall and F1. *Starts after: T-12*	—	Build	Rushiraj Navar	Baseline numbers printed for the first labelled pairs
Shopper interviews (3) 20 minutes each; notes and a one-page summary. Rushiraj and Jenish take one each. *Starts at the start of the sprint*	—	Research and process	Om Pandya	Notes in #elicitation

Data & Evaluation Engineer

Task	Story	Type	Owner	Done when
T-15 Elicitation write-up Survey counts, interviews, observation, expert interview and API limits written into docs/ and Lab 6 §5. *Starts after: survey closed, interviews done*	—	Build	Harshkumar Kodavala	Findings committed and posted in #elicitation
Observe one real purchase Think-aloud: sources used, order, where they gave up. *Starts at the start of the sprint*	—	Research and process	Harshkumar Kodavala	Notes in #elicitation

QA Engineer

Task	Story	Type	Owner	Done when
T-02 Repository, CI and secret scanning UTF-8 contract export fix, LF line endings, Windows CI job, secret scan, branch protection. *Starts after: T-01*	US-25	Build	Palak Patel	CI green on Linux and Windows; ₹ prints correctly; no secret in the repo
T-16 Sprint 1 review check Run every exit criterion and the live drill (one LLM key off); report the gaps. *Starts after: T-03 – T-15*	—	Build	Palak Patel	Checklist report ready before the review

Other planned work

Surveys and interviews, the concept poster, API sign-ups and smoke tests (including Reddit's and YouTube's terms), and the ML track's dataset and Colab setup.

Testing focus

A contract test for every stage, unit tests for intake and the interview, and CI running lint, types and tests.

Sprint review

- **Demonstration.** Type a request in the browser or the CLI and get a rough ranked list end to end; show the detected intent and criteria; answer the chat interview.
- **Live failure drill.** Switch off one LLM provider key live; the search still finishes.

Exit criteria

Sprint 1 is complete only when all of the following are true:

1. The CLI and the browser both run a request through every stage and show a ranked list with sources, even if rough
2. Every stage has a contract and a contract test, and swapping any stage for a failing one triggers its fallback
3. Detection shows category, use, budget, region and rules; “watch” triggers a question; with every LLM switched off, the keyword fallback still detects the category and budget
4. Criteria with definitions appear for three categories, including one the app has never seen, and the fallback set appears when generation fails
5. The interview chat runs at least 5 questions and stores the answers
6. Switching off one LLM provider key does not stop the app

7. CI runs lint, types and tests on every pull request; no secret is in the repo; every one of the 10 members has at least one merged pull request under their own name
8. The survey has 30 or more responses, and the interviews, observation, expert interview and API limits are written up
9. The poster is printed

5. Sprint 2: Research and the safety net

Planned dates	5 – 18 October 2026
Sprint review	Saturday 17 October 2026
Scrum Master	Yashkumar Patel
Story points	37
Sprint goal	Real research: sources fetched in parallel, claims extracted with exact quotes, retrieved text treated as untrusted, and the core logic tested offline.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-08	Understand rough answers like "medium"	Must	3	Pal Jalodara, with Manu Pardhan
US-12	Research several sources in parallel	Must	8	Jenish Kavad
US-13	Score Reddit quality and count recommenders exactly	Should	3	Vaishvi Mehta
US-14	Extract claims with exact quotes	Must	8	Jhanvi Barot
US-27	Fetch sources in parallel	Must	2	Jenish Kavad
US-28	Treat retrieved content as untrusted	Must	3	Pal Jalodara
US-32	Core logic testable offline	Must	3	Harshkumar Kodavala
US-46	Circuit breaker and provider rotation	Must	4	Jhanvi Barot
US-48	Golden tests for broken LLM output	Must	3	Palak Patel
	Total		**37**	

Work by role

AI Engineer

Task	Story	Type	Owner	Done when
T-17 Circuit breaker and provider rotation Trip a provider after repeated failures, cool-downs, round-robin pool for bulk extraction, daily token counters. *Starts after: T-03*	US-46	Build	Jhanvi Barot	Fault tests show rotation and cool-down; calls stop before a budget runs out
T-22 Extract claims with exact quotes Per-source extraction of products, claims and opinions; quotes checked against the source; experience versus spec typing. *Starts after: T-17, T-21*	US-14	Build	Jhanvi Barot	Every claim has a URL, quote and time; invalid output is retried once, then dropped
Velocity reset and Sprint 3 refinement Re-set velocity from Sprints 1–2; move Plus stories out if needed. Om Pandya runs refinement as Product Owner. *Starts after: the Sprint 2 review*	—	Research and process	Jhanvi Barot	Sprint 3 backlog agreed

Backend Engineer

Task	Story	Type	Owner	Done when
T-19 Research several sources 4–6 queries built from the rubric; Reddit, Serper and review-site adapters; near-duplicate merge; URL and fetch time stored; usage counters. *Starts after: T-06, T-18*	US-12	Build	Jenish Kavad	3+ source types per search; one failing source never stops the run
T-20 Fetch sources in parallel Fixed-size thread pool and per-API rate limits. *Starts after: T-19*	US-27	Build	Jenish Kavad	Total fetch time is close to the slowest source
T-21 Treat retrieved content as untrusted Strip HTML, cap length, pass text as labelled data, accept only schema JSON; 10 hostile test pages. *Starts after: T-19*	US-28	Build	Pal Jalodara	The hostile pages change no ranking
T-23 Reddit quality and exact recommender counts Thread and comment scores, comment trees 3 levels deep, Aho-Corasick counting with one count per user, 180-day half-life. *Starts after: T-19*	US-13	Build	Vaishvi Mehta	The same threads always give the same counts; “1000XM5” inside “WF-1000XM5” is not counted
T-24 Understand rough answers One mapping table (low 2, medium 5, high 8, good enough 3) used everywhere, plus a switch to exact values. *Starts after: T-06*	US-08	Build	Pal Jalodara	Unit tests cover every phrase in the table

Frontend Engineer

Task	Story	Type	Owner	Done when
T-26 Research progress design Live stage-by-stage progress with sources found per type. *Starts after: T-09*	—	Screen design	Yashkumar Patel	Design approved in #dev
T-27 Research progress build and connection Progress stream from the pipeline shown while a search runs. *Starts after: T-19, T-26*	—	Build	Yashkumar Patel	Progress moves stage by stage in a real search

Frontend Engineer & Testing

Task	Story	Type	Owner	Done when
T-28 Rough-answer chips in the chat Low, medium, high and good-enough buttons plus an exact-value switch. *Starts after: T-11, T-24*	US-08	Build	Manu Pardhan	Rough and exact answers both reach the API correctly

ML Engineer

Task	Story	Type	Owner	Done when
T-29 Matcher training data Candidate pairs pulled from recorded runs, a hard-negative miner, and a Colab fine-tune notebook (v0). *Starts after: T-13, T-19*	US-15	Model training	Om Pandya	300+ labelled pairs; v0 model scored against the fuzzy baseline
T-30 Seeded conflicts and trust eval runner 30 seeded conflicts with known answers, and baselines for majority vote and tier order. *Starts after: T-13*	US-16	Build	Rushiraj Navar	Baseline accuracy printed for the 30 conflicts
Labelling sessions 30 minutes a day in pairs; kappa ≥ 0.8 on 50 shared pairs. Rushiraj Navar labels too. *Starts after: T-12*	—	Research and process	Om Pandya	Kappa recorded in ml/DATA.md

Data & Evaluation Engineer

Task	Story	Type	Owner	Done when
T-18 Core logic testable offline Recorded API response fixtures, a capped recorder script, coverage gates of 70% on core and 80% on decide/. *Starts after: T-02*	US-32	Build	Harshkumar Kodavala	Fast suite runs with no network in under 60 s

QA Engineer

Task	Story	Type	Owner	Done when
T-25 Golden tests for broken LLM output About 20 malformed outputs per parser: fenced JSON, missing keys, wrong types, cut-off text, embedded instructions. *Starts after: T-22*	US-48	Build	Palak Patel	Every case gives the canonical shape or a clean fallback, never a crash
T-31 Sprint 2 review check Exit checks plus a drill: switch Reddit off live. *Starts after: T-17 – T-30*	—	Build	Palak Patel	Checklist report ready before the review

Other planned work

Matching pairs (public sets plus at least 150 of our own), the first 15 seeded conflicts, the 50 simulated shoppers, the first recorded real runs, the Lab 6 §5 rewrite with survey evidence, and a velocity re-set.

Testing focus

Golden tests for every parser, recorded API responses, and coverage of 70% or more on core modules.

Sprint review

- **Demonstration.** Live research across Reddit, search and review sites; extracted claims with quotes and exact mention counts; a hostile test page that changes nothing.
- **Live failure drill.** Rate-limit a provider; the circuit breaker moves on without the user noticing.

6. Sprint 3: Trustworthy end to end

Planned dates	19 October – 1 November 2026
Sprint review	Saturday 31 October 2026
Scrum Master	Vaishvi Mehta
Story points	40
Sprint goal	Trustworthy ranked results: learned product matching and source trust, hard rules and exclusions, ranking with coverage, and every claim clickable to its source.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-09	Keep must-haves separate from preferences	Must	5	Pal Jalodara, with Manu Pardhan
US-10	Rule out brands and features	Must	3	Jenish Kavad, with Manu Pardhan
US-15	Match product names and variants	Must	8	Om Pandya, with Rushiraj Navar
US-16	Learned source trust, disputes and unknowns	Must	8	Rushiraj Navar, with Om Pandya
US-17	Filter out products that break the rules	Must	3	Vaishvi Mehta
US-18	Rank on the user's priorities with coverage	Must	8	Vaishvi Mehta
US-21	Click any claim to see its source	Must	3	Yashkumar Patel, with Jenish Kavad
US-40	Never recommend an unsourced product	Must	2	Jhanvi Barot
	Total		**40**	

Work by role

AI Engineer

Task	Story	Type	Owner	Done when
T-37 Never recommend an unsourced product Drop products with no source mention before scoring; log the reason. *Starts after: T-22*	US-40	Build	Jhanvi Barot	Planted fake product names never reach the results

Backend Engineer

Task	Story	Type	Owner	Done when
T-32 Must-haves separate from preferences Rubric with hard rules and weighted preferences that add up to 1, category defaults, and manual edits that survive a rebuild. *Starts after: T-24*	US-09	Build	Pal Jalodara	Hard rules only filter and preferences only reorder (unit tests)
T-33 Rule out brands and features Parse “no Xiaomi” or “not touchscreen” into exclusions; an “always” flag saves them for memory. *Starts after: T-32*	US-10	Build	Jenish Kavad	Excluded items never appear and show in the filtered-out list with the reason

Task	Story	Type	Owner	Done when
T-38 Filter products that break the rules Budget, hard rules and exclusions applied in code: shown, filtered out with a reason, or can't tell. *Starts after: T-32, T-33*	US-17	Build	Vaishvi Mehta	Zero recommended products break a hard rule on every test brief
T-39 Rank on the user's priorities with coverage Weighted sum, coverage %, unknowns estimated from similar products and marked, mention counts shown as evidence only. *Starts after: T-38, T-36*	US-18	Build	Vaishvi Mehta	The most-mentioned product does not win when it fits worst
T-40 Evidence endpoint API that returns source name, type, date and quote for any claim, and all sources for a Disputed claim. *Starts after: T-36*	US-21	Build	Jenish Kavad	Contract test for every claim in the example result

Frontend Engineer

Task	Story	Type	Owner	Done when
T-41 Results page design Ranked cards, coverage, Disputed, Unknown and Estimated badges, filtered-out list, source drawer. *Starts after: T-27*	—	Screen design	Yashkumar Patel	Design approved in #dev
T-42 Results page build and claim sources Rebuild the results page and connect it; clicking any claim opens its source and exact quote. *Starts after: T-39, T-40, T-41*	US-21	Build	Yashkumar Patel	Every factual claim on the page opens its source

Frontend Engineer & Testing

Task	Story	Type	Owner	Done when
T-43 Rubric editor design Hard-rule toggles, weight controls, exclusion chips. *Starts after: T-28*	US-09	Screen design	Manu Pardhan	Design approved in #dev
T-44 Rubric editor build and connection Users see which statements became hard rules and can switch them; exclusions are editable. *Starts after: T-32, T-33, T-43*	US-09	Build	Manu Pardhan	Edits survive a rubric rebuild

ML Engineer

Task	Story	Type	Owner	Done when
T-34 Matcher training (final) Train on the final labelled pairs, pick the threshold for precision ≥ 0.95, and upload the weights to Hugging Face pinned by revision. *Starts after: T-29*	US-15	Model training	Om Pandya	Eval report beats the fuzzy baseline on the held-out pairs

Task	Story	Type	Owner	Done when
T-35 Product matcher in the match stage Normalise, block, cross-encoder score, threshold, union-find with a chaining guard, variant rules; fuzzy matching stays as the fallback. *Starts after: T-34*	US-15	Build	Om Pandya	method = model when the model ran and fuzzy on fallback; under 5 s
T-36 Learned source trust Site trust table including Indian review sites, affiliate rule, iterative truth discovery with Beta smoothing, and Disputed and Unknown states. *Starts after: T-30, T-22*	US-16	Build	Rushiraj Navar	Beats majority vote on the seeded conflicts; trust_method reported
Interview simulator prep 50 synthetic shoppers with hidden weights, ready for US-04. *Starts after: T-32*	—	Research and process	Om Pandya	Simulator script runs the fixed-order baseline

Data & Evaluation Engineer

Task	Story	Type	Owner	Done when
T-45 Sprint 3 baselines Run every eval on the recorded runs and set the floors in thresholds.yaml (the plan fixes final targets here). *Starts after: T-35, T-36, T-39*	—	Build	Harshkumar Kodavala	Baseline report committed; targets confirmed with the team

QA Engineer

Task	Story	Type	Owner	Done when
T-46 Sprint 3 review check Exit checks plus a drill: force the matcher to fail and show the fuzzy fallback. *Starts after: T-32 – T-45*	—	Build	Palak Patel	Checklist report ready before the review
Weekly citation audit 10 citations checked by hand each week from now on. *Starts after: T-42*	—	Research and process	Palak Patel	Audit log in docs/

Other planned work

Train the matcher on Colab, finish the 250 own pairs and 30 conflicts, record every baseline number, and run the full flow on 5 briefs.

Testing focus

Zero hard-rule violations on every brief, matcher and trust evaluation scripts, and an invented-product test.

Sprint review

- **Demonstration.** Request to ranked, sourced shortlist; the matcher against fuzzy matching; trust against majority voting; Disputed and Unknown labels; the filtered-out list; a planted invented product dropped.
- **Live failure drill.** Switch Reddit off; the result arrives with a lower coverage note.

7. Sprint 4: Smarter and safer

Planned dates	2 – 15 November 2026
Sprint review	Saturday 14 November 2026
Scrum Master	Harshkumar Kodavala
Story points	36
Sprint goal	Smarter and honest: an adaptive interview, confidence and ties from simulations, explanations, login with private data, graceful failures and a CI quality gate.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-04	Smart interview that asks only what matters	Should	8	Om Pandya, with Jenish Kavad
US-19	Honest confidence and ties from simulations	Should	5	Rushiraj Navar, with Om Pandya and Manu Pardhan
US-20	Explain why a product fits	Must	5	Jhanvi Barot, with Manu Pardhan
US-30	Login and private preferences	Must	5	Harshkumar Kodavala, with Vaishvi Mehta and Yashkumar Patel
US-31	Handle outside failures gracefully	Must	5	Pal Jalodara
US-39	Understand questions, skips and off-topic replies in the interview	Must	3	Jhanvi Barot, with Manu Pardhan
US-49	Recorded-run replay and the CI quality gate	Must	5	Harshkumar Kodavala
	Total		**36**	

Work by role

AI Engineer

Task	Story	Type	Owner	Done when
T-50 Explain why a product fits Top 3 criteria by contribution computed in code, one trade-off, unverified facts named; the LLM only writes the words; template fallback. *Starts after: T-39*	US-20	Build	Jhanvi Barot	Every explanation number matches the scoring code
T-51 Questions, skips and off-topic replies Classify replies into 7 types and handle each; reply text is data, never instructions. *Starts after: T-47*	US-39	Build	Jhanvi Barot	90%+ correct on 70 sample replies

Backend Engineer

Task	Story	Type	Owner	Done when
T-48 Interview API integration Serve the information-gain order through the interview API; pass uncertainty into the rubric. *Starts after: T-47*	US-04	Build	Jenish Kavad	The chat runs the adaptive order end to end
T-52 User tables and row-level security Migrations for users, preferences and searches with RLS policies, and a down-migration. *Starts after: T-14*	US-30	Build	Vaishvi Mehta	User A cannot read user B's rows (DB test)
T-54 Handle outside failures gracefully Timeouts from the contract table, one retry, lower-coverage notes, a clear message and retry when every LLM fails. *Starts after: T-17*	US-31	Build	Pal Jalodara	Each fault test ends in a result or a clear message, never a crash

Frontend Engineer

Task	Story	Type	Owner	Done when
T-56 Login design Sign-in, signed-in header, guest notice. *Starts after: T-42*	US-30	Screen design	Yashkumar Patel	Design approved in #dev
T-57 Login build and connection Google sign-in through Supabase; guest mode still works. *Starts after: T-53, T-56*	US-30	Build	Yashkumar Patel	Sign in, sign out and guest flows pass a Playwright test

Frontend Engineer & Testing

Task	Story	Type	Owner	Done when
T-58 Confidence, ties and explanations design Confidence label, "too close to call" card with the separating question, explanation paragraph, question-back replies in the chat. *Starts after: T-44*	US-19	Screen design	Manu Pardhan	Design approved in #dev
T-59 Confidence, ties and explanations build Show chance first, labels, ties, explanations and US-39 replies in the app. *Starts after: T-49, T-50, T-51, T-58*	US-20	Build	Manu Pardhan	Every state in the example result renders

ML Engineer

Task	Story	Type	Owner	Done when
T-47 Bayesian adaptive interview Normal belief per weight, conjugate update, importance × uncertainty question choice, stopping rules; fixed order stays as the fallback. *Starts after: T-32, interview simulator*	US-04	Build	Om Pandya	30% fewer questions on 50 simulated shoppers; ordering field reported

Task	Story	Type	Owner	Done when
T-49 Confidence and ties from simulations 1,000 seeded simulations, chance_first, high/medium/low labels, ties with the separating criterion; the weighted score stays as the fallback. *Starts after: T-39, T-47*	US-19	Build	Rushiraj Navar	Same seed gives the same output; Spearman ≥ 0.9; under 2 s

Data & Evaluation Engineer

Task	Story	Type	Owner	Done when
T-53 Login and private preferences Supabase Auth with Google sign-in; every API route scoped to the signed-in user. *Starts after: T-52*	US-30	Build	Harshkumar Kodavala	API test: user A cannot read user B's data
T-55 Recorded-run replay and CI quality gate 5 recorded real runs replayed on every PR; floors in thresholds.yaml; fail on a drop of more than 10 points. *Starts after: T-45*	US-49	Build	Harshkumar Kodavala	A deliberately broken PR is blocked by the gate

QA Engineer

Task	Story	Type	Owner	Done when
T-60 Sprint 4 review check Exit checks plus a drill: switch the database off live. *Starts after: T-47 – T-59*	—	Build	Palak Patel	Checklist report ready before the review

Other planned work

Run the interview simulation and the rank-stability test.

Testing focus

Recorded real-run replay and the CI quality gate go live; fault tests for every source and provider.

Sprint review

- **Demonstration.** The shorter interview against the fixed questionnaire with numbers; a question asked mid-interview handled correctly; ties flagged; explanations; login isolation.
- **Live failure drill.** Switch every LLM provider off; the user gets a cached result or a clear message, never a crash.

8. Sprint 5: Feature-complete

Planned dates	16 – 29 November 2026
Sprint review	Saturday 28 November 2026
Scrum Master	Pal Jalodara
Story points	41 (36 core + 5 Plus)
Sprint goal	Personal and complete: preference memory, a citation check, compare and instant re-ranking, prices and buy links, YouTube and hard pages, and missing-fact lookup.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-06	Remember preferences across searches	Should	8	Om Pandya, with Vaishvi Mehta and Manu Pardhan
US-22	Check every explanation sentence against its evidence	Should	5	Rushiraj Navar, with Jhanvi Barot
US-23	Compare products and re-rank instantly	Must	8	Yashkumar Patel
US-24	Current price and buy link for the exact model	Must	5	Jenish Kavad, with Manu Pardhan
US-38	Add YouTube and hard-to-read pages	Must	5	Pal Jalodara
US-41	Look up missing facts for the top products	Must	5	Jhanvi Barot
US-42	Read community posts like a person would (Plus)	Should	5	Harshkumar Kodavala
	Total		**41**	

Work by role

AI Engineer

Task	Story	Type	Owner	Done when
T-66 Look up missing facts for the top products Targeted lookups for top-5 gaps, at most 6 searches and 1 batched LLM call, 7-day cache, cited values only. *Starts after: T-39*	US-41	Build	Jhanvi Barot	Coverage before and after reported on 10 briefs

Backend Engineer

Task	Story	Type	Owner	Done when
T-61 pgvector memory table pgvector extension, vector(384) column, RLS, optional HNSW index. *Starts after: T-52*	US-06	Build	Vaishvi Mehta	Migration runs up and down; RLS test passes
T-64 Current price and buy link Serper shopping results, parallel retailer lookups, title must match the model, as-of time, 24-hour staleness label. *Starts after: T-35*	US-24	Build	Jenish Kavad	Every price shows an as-of time; mismatched links are rejected

Task	Story	Type	Owner	Done when
T-65 YouTube and hard-to-read pages YouTube Data API (3–5 videos), transcripts only if the terms allow, Jina Reader for JavaScript pages. *Starts after: T-19, T-21*	US-38	Build	Pal Jalodara	With YouTube or Jina down, the search continues with a coverage note

Frontend Engineer

Task	Story	Type	Owner	Done when
T-68 Compare and re-rank design Side-by-side compare (up to 4) and weight sliders with number boxes. *Starts after: T-57*	US-23	Screen design	Yashkumar Patel	Design approved in #dev
T-69 Compare and re-rank instantly Client-side scoring that matches the backend formula, sliders that re-rank in under 1 s with no server call, rank movement shown. *Starts after: T-39, T-68*	US-23	Build	Yashkumar Patel	Parity test: browser and backend scores agree

Frontend Engineer & Testing

Task	Story	Type	Owner	Done when
T-70 Remembered answers and price cards design “Remembered” pre-filled answers in the chat; price, store and buy-link cards. *Starts after: T-59*	US-06	Screen design	Manu Pardhan	Design approved in #dev
T-71 Remembered answers and price cards build Pre-filled answers can be changed in one click; prices show as-of times and a “may have changed” label. *Starts after: T-62, T-64, T-70*	US-24	Build	Manu Pardhan	Both flows run end to end in the browser

ML Engineer

Task	Story	Type	Owner	Done when
T-62 Remember preferences across searches Local MiniLM embeddings, top 5 above 0.7 similarity, 90-day decay, category guard; empty list on failure. *Starts after: T-61, T-53*	US-06	Build	Om Pandya	50% fewer repeated questions on 20 seeded sessions; zero cross-user leaks
T-63 Check explanation sentences against evidence NLI entailment on cited quotes, chunk max, number rule, threshold tuned on 40 and tested on 60 sentences. *Starts after: T-50*	US-22	Build	Rushiraj Navar	Catches 80%+ of wrong claims; rejects at most 15% of right ones

Data & Evaluation Engineer

Task	Story	Type	Owner	Done when
T-67 Read community posts like a person would Negation, sarcasm markers, backed complaints, cross-community splits. Plus story: moves out first if late. *Starts after: T-23*	US-42	Build	Harshkumar Kodavala	Passes 40 labelled sentences and 10 known community splits

QA Engineer

Task	Story	Type	Owner	Done when
T-72 Sprint 5 review check Exit checks plus a drill: switch YouTube and one LLM key off. *Starts after: T-61 – T-71*	—	Build	Palak Patel	Checklist report ready before the review
Usability tests 5 non-technical users; Manu Pardhan helps run them. *Starts after: T-69*	—	Research and process	Palak Patel	Findings in docs/ and new backlog items

Other planned work

Write the 100 claim and evidence pairs and run the citation test, run the memory test, run usability tests with 5 non-technical users, and do the first run of the 20-brief study.

Testing focus

Citation-check evaluation, enrichment before and after, and fallbacks for the NLI model and embeddings.

Sprint review

- **Demonstration.** Memory pre-fill; enrichment raising coverage; planted wrong claims caught; sliders and compare; prices with times; a YouTube source.
- **Live failure drill.** Disable the NLI model; explanations fall back to quoted sentences. The product is feature-complete after this review.

9. Sprint 6: Worst-case proof and presentation

Planned dates	30 November – 13 December 2026
Sprint review	Saturday 12 December 2026
Scrum Master	Palak Patel
Story points	49 (31 core + 10 Plus + 8 Stretch)
Sprint goal	Polish, prove and demo: accessibility, speed and cache, traceability, the worst-case suite, the evaluation study, then Plus and stretch work only if time allows.

User stories in this sprint

ID	Title	Priority	Points	Owner
US-07	Explain unfamiliar terms on request	Should	2	Manu Pardhan
US-11	See and control what the app remembers	Should	3	Vaishvi Mehta, with Yashkumar Patel
US-29	Trace every claim to its evidence	Must	3	Jenish Kavad
US-33	Plain language and accessible controls	Must	5	Manu Pardhan
US-34	Fast responses and exact-match cache	Should	5	Pal Jalodara
US-35	Modules with clear interfaces	Should	3	Pal Jalodara
US-36	Classifier for promotional and duplicate comments (Stretch)	Could	8	Om Pandya, with Rushiraj Navar
US-37	Save or share results (Plus)	Could	3	Yashkumar Patel
US-47	Per-user rate limits (Plus)	Should	2	Vaishvi Mehta
US-50	Worst-case test suite	Must	5	Palak Patel
US-51	Measure consistency, personalisation and popularity bias	Should	5	Harshkumar Kodavala
US-53	Progress survives a dropped connection (Plus)	Could	2	Jenish Kavad
US-54	Team diagnostics page (Plus)	Could	3	Jhanvi Barot
	Total		**49**	

Work by role

AI Engineer

Task	Story	Type	Owner	Done when
T-88 Team diagnostics page Provider status, recent LLM calls, stage times, fallbacks and coverage, behind team login. *Starts after: T-53*	US-54	Build	Jhanvi Barot	Hidden from non-team users
Demo rehearsal Two full runs of the 9-minute demo with the live drill, plus a recorded backup run. Whole team. *Starts after: T-89*	—	Research and process	Jhanvi Barot	Demo timed under 10 minutes

Backend Engineer

Task	Story	Type	Owner	Done when
T-74 Memory control API View, edit, export as JSON and delete preferences (with their embeddings); account delete removes everything. *Starts after: T-62*	US-11	Build	Vaishvi Mehta	Delete tests confirm no rows or vectors remain
T-77 Trace every claim to its evidence Store each claim and ranking decision with source ID, quote and time; saved results reopen with the same evidence. *Starts after: T-40*	US-29	Build	Jenish Kavad	A random-claim test finds every source in the database
T-79 Fast responses and exact-match cache 24-hour exact-match cache, interview turns within 5 s, searches within 120 s. *Starts after: T-65*	US-34	Build	Pal Jalodara	A repeat search returns in 3 s or less
T-80 Modules with clear interfaces Provider swap by config only, documented module inputs and outputs, one integration test per boundary. *Starts after: T-79*	US-35	Build	Pal Jalodara	A new category runs end to end with no code change
T-86 Per-user rate limits 10 searches a minute per user or address; 200 requests a minute overall. *Starts after: T-53*	US-47	Build	Vaishvi Mehta	Over the limit, the user sees when to try again
T-87 Progress survives a dropped connection Reconnect with increasing waits; resume from the last finished stage. *Starts after: T-27*	US-53	Build	Jenish Kavad	Killing the connection mid-search recovers the progress

Frontend Engineer

Task	Story	Type	Owner	Done when
T-75 Memory settings page design List of remembered preferences with edit, export and delete. *Starts after: T-69*	US-11	Screen design	Yashkumar Patel	Design approved in #dev
T-76 Memory settings page build Connect the page to the memory control API. *Starts after: T-74, T-75*	US-11	Build	Yashkumar Patel	View, edit, export and delete work in the browser
T-85 Save or share results PDF and CSV export and a read-only share link. *Starts after: T-77*	US-37	Build	Yashkumar Patel	A share link opens the same results without an account

Frontend Engineer & Testing

Task	Story	Type	Owner	Done when
T-73 Explain unfamiliar terms "What does this mean?" link using the saved criterion definitions; does not count as an answer. *Starts after: T-05, T-59*	US-07	Build	Manu Pardhan	Every question with a technical term has the link
T-78 Plain language and accessible controls Keyboard-only flows, WCAG AA contrast, number boxes on sliders, Grade 8 wording. *Starts after: T-69*	US-33	Build	Manu Pardhan	Axe check passes; 4 of 5 testers understand a result unaided

ML Engineer

Task	Story	Type	Owner	Done when
T-84 Promotional comment classifier (stretch) Only if the Sprint 4–5 ML targets are met. Rushiraj Navar co-owns it. *Starts after: T-23*	US-36	Build	Om Pandya	Beats the US-13 rules on F1; flagged comments down-weighted, never deleted
Final report and evaluation chapter Updated Lab 6 report and the ML chapter built from the eval reports. Harshkumar Kodavala writes the evaluation report. *Starts after: T-83*	—	Research and process	Om Pandya	Submitted with the ZIP

Data & Evaluation Engineer

Task	Story	Type	Owner	Done when
T-82 Consistency, personalisation and popularity bias Paraphrase pairs, 5 personas, popularity scenarios, all in the metrics report. *Starts after: T-55*	US-51	Build	Harshkumar Kodavala	8 of 10 paraphrase pairs agree; 3+ different picks for 5 personas
T-83 20-brief study 20 briefs across 4 categories; 3 members write their own top 3s; ShopSense compared with a plain AI answer. *Starts after: T-82*	—	Build	Harshkumar Kodavala	Overlap and unsupported-claim counts in the evaluation report

QA Engineer

Task	Story	Type	Owner	Done when
T-81 Worst-case test suite One fault test per worst-case row, a Playwright test on every merge to main, a written drill script. *Starts after: T-54*	US-50	Build	Palak Patel	Every worst-case row has a passing test
T-89 Final release check Full demo script dry run, all seven test layers green, every member with merged PRs. *Starts after: T-73 – T-88*	—	Build	Palak Patel	Release report ready before the final review

Other planned work

Final evaluation tables, the final 20-brief study, the Lab 6 report update, the poster update and two demo rehearsals. Feature freeze Monday 7 December; code freeze Wednesday 9 December.

Testing focus

One fault test per row of the worst-case table, the Playwright browser test, and the consistency, personalisation and popularity-bias metrics.

Sprint review

- **Demonstration.** The full demo script, including a live drill, then the results slide.
- **Live failure drill.** Break a key, a source and the database in turn using the written drill script.

10. Work summary by member

Number of tasks each member owns in each sprint, including tasks done by hand, and the story points they own. Every member owns tasks in every sprint except Rushiraj Navar in Sprint 6, where he co-owns the Stretch story US-36 with Om Pandya.

Member	Role	S1	S2	S3	S4	S5	S6	Tasks	Points
Om Pandya	ML Engineer	2	2	3	1	1	2	11	32
Rushiraj Navar	ML Engineer	1	1	1	1	1	0	5	18
Jhanvi Barot	AI Engineer	5	3	1	2	1	2	14	45
Harshkumar Kodavala	Data & Evaluation Engineer	2	1	1	2	1	2	9	23
Palak Patel	QA Engineer	2	2	2	1	2	2	11	11
Pal Jalodara	Backend Engineer	2	2	1	1	1	2	9	34
Vaishvi Mehta	Backend Engineer	2	1	2	1	1	2	9	19
Jenish Kavad	Backend Engineer	2	2	2	1	1	2	10	27
Yashkumar Patel	Frontend Engineer	3	2	2	2	2	3	14	19
Manu Pardhan	Frontend Engineer & Testing	2	1	2	2	2	2	11	11
Total		**23**	**17**	**17**	**14**	**13**	**19**	**103**	**239**